

Experiment 07 – Database Normalisation

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Aim

To understand 1NF, 2NF, 3NF and BCNF, and to practice identifying candidate keys and the highest normal form satisfied by a relation given its functional dependencies.

Objective

- Determine the candidate key(s) of a relation from its functional dependencies.
- Check a relation for partial and transitive dependencies.
- Classify each relation into 1NF, 2NF, 3NF, or BCNF.
- Apply the theory to a set of worked problems.

Find the Highest Normal Form of Each Relation

Problem 1: R(A, B, C, D, E)

$CE \rightarrow D$
 $D \rightarrow B$
 $C \rightarrow A$

Candidate key: CE

- CE is the only candidate key — closure(CE) reaches every attribute, and neither C nor E alone does.
- C, a proper subset of the key CE, determines A (a non-prime attribute) → partial dependency → fails 2NF.
- $D \rightarrow B$ is a non-prime attribute determining another non-prime attribute → would also fail 3NF.
- $D \rightarrow B$ also has a non-superkey determinant → would fail BCNF.

Highest normal form: 1NF (fails 2NF due to the partial dependency $C \rightarrow A$)

Problem 2: R(A, B, C, D, E, F)

$AB \rightarrow C$
 $DC \rightarrow AE$
 $E \rightarrow F$

Candidate key: ABD

- ABD is the candidate key — its closure reaches all six attributes, and no two-attribute subset of it does.
- AB, a proper subset of ABD, determines C (non-prime) → partial dependency → fails 2NF.

Highest normal form: 1NF (fails 2NF due to the partial dependency $AB \rightarrow C$)

Problem 3: R(A, B, C, D, E, F, G, H, I)

$AB \rightarrow C$
 $BD \rightarrow EF$
 $AD \rightarrow GH$
 $A \rightarrow I$

Candidate key: ABD

- ABD is the candidate key — closure(ABD) reaches all nine attributes; no pair subset does.
- AB, a proper subset of ABD, determines C (non-prime) → partial dependency → fails 2NF.

Highest normal form: 1NF (fails 2NF due to the partial dependency $AB \rightarrow C$)

Problem 4: R(A, B, C, D, E)

$AB \rightarrow CD$
 $D \rightarrow A$

$BC \rightarrow DE$

Candidate key: AB and BC (both are candidate keys)

- closure(AB): $AB \rightarrow CD$ gives C, D; $D \rightarrow A$ adds nothing new; $BC \rightarrow DE$ (B, C now present) adds E $\rightarrow$ full set.
- closure(BC): $BC \rightarrow DE$ gives D, E; $D \rightarrow A$ adds A $\rightarrow$ full set.
- So both AB and BC are minimal candidate keys — prime attributes are A, B, C; non-prime are D, E.
- No proper subset of AB or BC (A, B or C alone) determines a non-prime attribute $\rightarrow$ no partial dependency $\rightarrow$ 2NF holds.
- $D \rightarrow A$: D is not a superkey, but A is a prime attribute, which 3NF explicitly allows $\rightarrow$ 3NF holds.
- $D \rightarrow A$ still violates BCNF, since D alone is not a superkey.

Highest normal form: 3NF (holds 3NF but not BCNF, because of $D \rightarrow A$)

Problem 5: R(A, B, C, D, E)

$BC \rightarrow ADE$
 $D \rightarrow B$

Candidate key: BC and CD (both are candidate keys)

- closure(BC): $BC \rightarrow ADE$ gives A, D, E directly $\rightarrow$ full set.
- closure(CD): $D \rightarrow B$ gives B; now $BC \rightarrow ADE$ (B, C present) gives A, E $\rightarrow$ full set.
- Prime attributes: B, C, D (each appears in some candidate key); non-prime: A, E.
- No proper subset of BC or CD determines a non-prime attribute $\rightarrow$ 2NF holds.
- $D \rightarrow B$: D is not a superkey, but B is prime, which 3NF allows $\rightarrow$ 3NF holds.
- $D \rightarrow B$ still violates BCNF, since D alone is not a superkey.

Highest normal form: 3NF (holds 3NF but not BCNF, because of $D \rightarrow B$)

Problem 6: R(A, B, C)

$A \rightarrow B$
 $B \rightarrow C$
 $C \rightarrow A$

Candidate key: A, B, and C are each individually a candidate key

- closure(A) = {A, B, C} via $A \rightarrow B \rightarrow C$. Likewise closure(B) and closure(C) each reach every attribute.
- Every attribute is therefore prime — there are no non-prime attributes to violate 2NF or 3NF with.
- Check BCNF: for $A \rightarrow B$, the determinant A is itself a candidate key, so it is a superkey — no violation. The same holds for $B \rightarrow C$ and $C \rightarrow A$.
- Since every functional dependency's left-hand side is a superkey, the BCNF condition is satisfied.

Highest normal form: BCNF (every determinant is itself a candidate key / superkey)

Problem 7: R(F1, F2, F3, F4, F5)

$F1 \rightarrow F3$
 $F2 \rightarrow F4$
 $(F1, F2) \rightarrow F5$

Candidate key: (F1, F2)

- closure(F1F2) reaches F3 (via F1), F4 (via F2) and F5 directly $\rightarrow$ full set; neither F1 nor F2 alone is sufficient.
- F1, a proper subset of the key, determines F3 (non-prime) $\rightarrow$ partial dependency $\rightarrow$ fails 2NF.
- F2, also a proper subset, determines F4 (non-prime) $\rightarrow$ a second partial dependency.

Highest normal form: 1NF (fails 2NF due to the partial dependencies $F1 \rightarrow F3$ and $F2 \rightarrow F4$)

Problem 8: R(A, B, C, D, E, F, G, H, I)

$AB \rightarrow C$
 $AD \rightarrow GH$

$BD \rightarrow EF$
 $A \rightarrow I$
 $H \rightarrow J$

Candidate key: ABD

- closure(ABD) reaches C, G, H, E, F and I through the four dependencies above — the full attribute set.
- AB, a proper subset of ABD, determines C (non-prime) $\rightarrow$ partial dependency $\rightarrow$ fails 2NF.

Highest normal form: 1NF (fails 2NF due to the partial dependency $AB \rightarrow C$)

Problem 9: R(A, B, C, D, E)

$A \rightarrow B$
 $B \rightarrow E$
 $C \rightarrow D$

Candidate key: AC

- closure(AC): $A \rightarrow B$ gives B, $B \rightarrow E$ gives E, $C \rightarrow D$ gives D $\rightarrow$ full set; A alone and C alone each fall short.
- A, a proper subset of AC, determines B (non-prime) $\rightarrow$ partial dependency $\rightarrow$ fails 2NF.
- C, also a proper subset, determines D (non-prime) $\rightarrow$ a second partial dependency.

Highest normal form: 1NF (fails 2NF due to the partial dependencies $A \rightarrow B$ and $C \rightarrow D$)

Learning Outcome

1. Understood functional dependencies and how to compute attribute closures.
2. Learned to identify candidate keys from a set of functional dependencies.
3. Understood the distinctions between 1NF, 2NF, 3NF, and BCNF.
4. Practiced classifying relations into their highest satisfied normal form.